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Quantum Mechanics
Lecture 6

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Recap

- When the potential is time independent, the Schrodinger Equation can be solved using the method of separation of variables.

$$\Psi(x,t) = \psi(x)f(t)$$

- Separable solutions provide **stationary states**, for which probability density and expectation values are time independent.

$$\Psi_n(x,t) = \psi_n(x)e^{-iE_nt/\hbar}$$

- Infinite potential well provides discrete energy Eigen values and Eigen states where the particles have definite energy and no uncertainty. The eigenstates are orthonormal.

$$E_n = \frac{\hbar^2 k_n^2}{2m} = \frac{n^2\pi^2\hbar^2}{2ma^2}.$$

$$\psi_n(x) = \sqrt{\frac{2}{a}}\sin\left(\frac{n\pi}{a}x\right).$$

$$n = 1, 2, 3...$$

- The generic solution is the superposition of all these eigenstate solutions with arbitrary coefficients.

$$\Psi(x,t) = \sum_{n=1}^{\infty} c_n \psi_n(x) e^{-iE_nt/\hbar}.$$

- The coefficients can be found out if the initial wave function of the system is known.

Further interpretation of the eigenstates

The eigenstates are defined by: $\hat{H}\psi_n = E_n\psi_n$

Also recall, that for the system: $\boxed{\Psi(x) = \sum_{n=1}^{\infty} c_n\psi_n(x)}$

Expectation of the Hamiltonian for the system

$$\langle H\rangle = \int \Psi^*\hat{H}\Psi dx \quad = \int dx\ \Psi^*\hat{H}\sum c_n\psi_n$$

$$= \int dx\ \Psi^*\sum c_n E_n\psi_n$$

$$= \sum |c_n|^2 E_n$$

Also, the normalisation for the system wave function:

$$\int \Psi^*(x)\Psi(x)\ dx = \sum |c_n|^2 = 1$$

- $|c_n|^2$ can be interpreted as the probability of finding the particle in $n-$the eigenstate with eigenvalue E_n.
- If the system is in eigenstate ψ_n, then the measurement will necessarily yield $\langle\hat{H}\rangle = E_n$
- If the system is in state $\Psi(x)$, which is a combination of the eigenstates, then the energy measurement determines the relative probabilities of the system being in eigenstates E_1, E_2.., so that, $\langle\hat{H}\rangle = \sum |c_n|^2 E_n$

Free particle Schrodinger Equation

For free particle, $V(x)=0$ everywhere.

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi,$$

This can be written as:

$$\frac{d^2\psi}{dx^2} = -k^2\psi,$$

$$\text{where } k \equiv \frac{\sqrt{2mE}}{\hbar}.$$

Simple Harmonic Equation

Solution:

$$\psi(x) = Ae^{ikx} + Be^{-ikx}.$$

Bringing the time dependent part,

$$\Psi(x,t) = \underbrace{Ae^{ik\left(x-\frac{\hbar k}{2m}t\right)}}_{\sim(x-vt)} + \underbrace{Be^{-ik\left(x+\frac{\hbar k}{2m}t\right)}}_{\sim(x+vt)}.$$

Indicates motion along positive or negative x direction with velocity v

Alternatively,

$$\Psi_k(x,t) = Ae^{i\left(kx-\frac{\hbar k^2}{2m}t\right)},$$

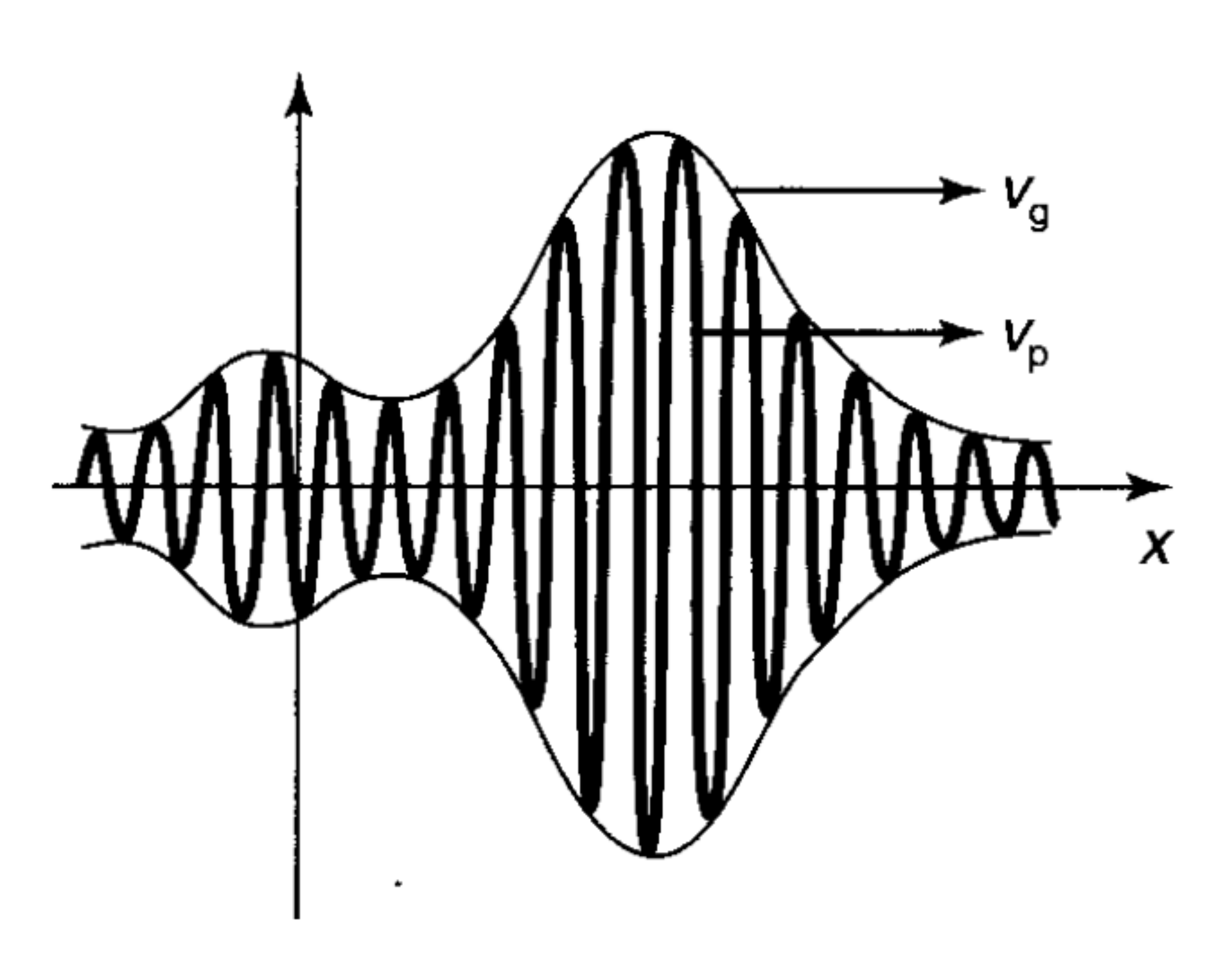
$$k \equiv \pm\frac{\sqrt{2mE}}{\hbar}, \quad \text{with } \begin{cases} k > 0 \Rightarrow & \text{traveling to the right,} \\ k < 0 \Rightarrow & \text{traveling to the left.} \end{cases}$$

Normalisation

Check:

$$\int_{-\infty}^{+\infty} \Psi_k^* \Psi_k \, dx = |A|^2 \int_{-\infty}^{+\infty} 1 \, dx = |A|^2(\infty).$$

The wave function is not normalisable !



In case of a free particle, then separable solution doesn't represent physically realisable states ! A free particle can't exist in a stationary state or to put it another way, there is no free particle with definite energy !

Then how to represent a free particle wave function ?

$$\Psi(x,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \phi(k) e^{i(kx - \frac{\hbar k^2}{2m} t)} \, dk.$$

Wave packet !

The normalisation now reads:

$$\int \Psi^*(x,t)\Psi(x,t)dx = \int \phi^*(p,t)\phi(p,t)dp = 1$$

Fourier transform

Or,

$$\Psi(x,0) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \phi(k) e^{ikx} \, dk. \quad \psi(x,0) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dp \, \phi(p) e^{ipx/\hbar}$$

Inverse Fourier transform

Momentum space wave function

$$\phi(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dx \, \psi(x,0) e^{-ipx/\hbar}$$

Example: Consider a **free particle wave function** localised within -a < x < a is released at t=0.

$$\Psi(x,0) = \begin{cases} A, & \text{if } -a < x < a, \\ 0, & \text{otherwise,} \end{cases}$$

where A and a are positive real constants. Find $\Psi(x,t)$.

This is not a stationary state, so we can't use

$$\Psi_n(x,t) = \psi_n(x)e^{-iE_nt/\hbar}$$

Solution: First we need to normalize $\Psi(x,0)$:

$$1 = \int_{-\infty}^{\infty} |\Psi(x,0)|^2 \, dx = |A|^2 \int_{-a}^{a} dx = 2a|A|^2 \Rightarrow A = \frac{1}{\sqrt{2a}}.$$

We need to use

$$\Psi(x,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \phi(k) e^{i(kx - \frac{\hbar k^2}{2m}t)} \, dk.$$

For that we need to find the momentum space wave function

Next we calculate $\phi(k)$, using Equation

$$\phi(k) = \frac{1}{\sqrt{2\pi}} \frac{1}{\sqrt{2a}} \int_{-a}^{a} e^{-ikx} \, dx = \frac{1}{2\sqrt{\pi a}} \left. \frac{e^{-ikx}}{-ik} \right|_{-a}^{a}$$

$$= \frac{1}{k\sqrt{\pi a}} \left(\frac{e^{ika} - e^{-ika}}{2i} \right) = \frac{1}{\sqrt{\pi a}} \frac{\sin(ka)}{k}.$$

$$\Psi(x,t) = \frac{1}{\pi\sqrt{2a}} \int_{-\infty}^{\infty} \frac{\sin(ka)}{k} e^{i(kx - \frac{\hbar k^2}{2m}t)} \, dk.$$

This integration has to be done numerically.

Two different types of solutions

We have studied two kinds of cases:

Free particle solution: The solution to the time-independent Schrodinger Equation provides non-normalisable states represented by continuous wave vectors k.

$$\Psi_k(x,t) = Ae^{i\left(kx - \frac{\hbar k^2}{2m}t\right)},$$

$$k \equiv \pm\frac{\sqrt{2mE}}{\hbar}, \quad \text{with} \begin{cases} k > 0 \Rightarrow & \text{traveling to the right,} \\ k < 0 \Rightarrow & \text{traveling to the left.} \end{cases}$$

$$\boxed{\Psi(x,t) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty}\phi(k)e^{i\left(kx - \frac{\hbar k^2}{2m}t\right)}\,dk.}$$

Wave packet

Infinite Square well potential: The solutions to the time independent Schrodinger Equations are normalisable and labelled by discrete quantum numbers n.

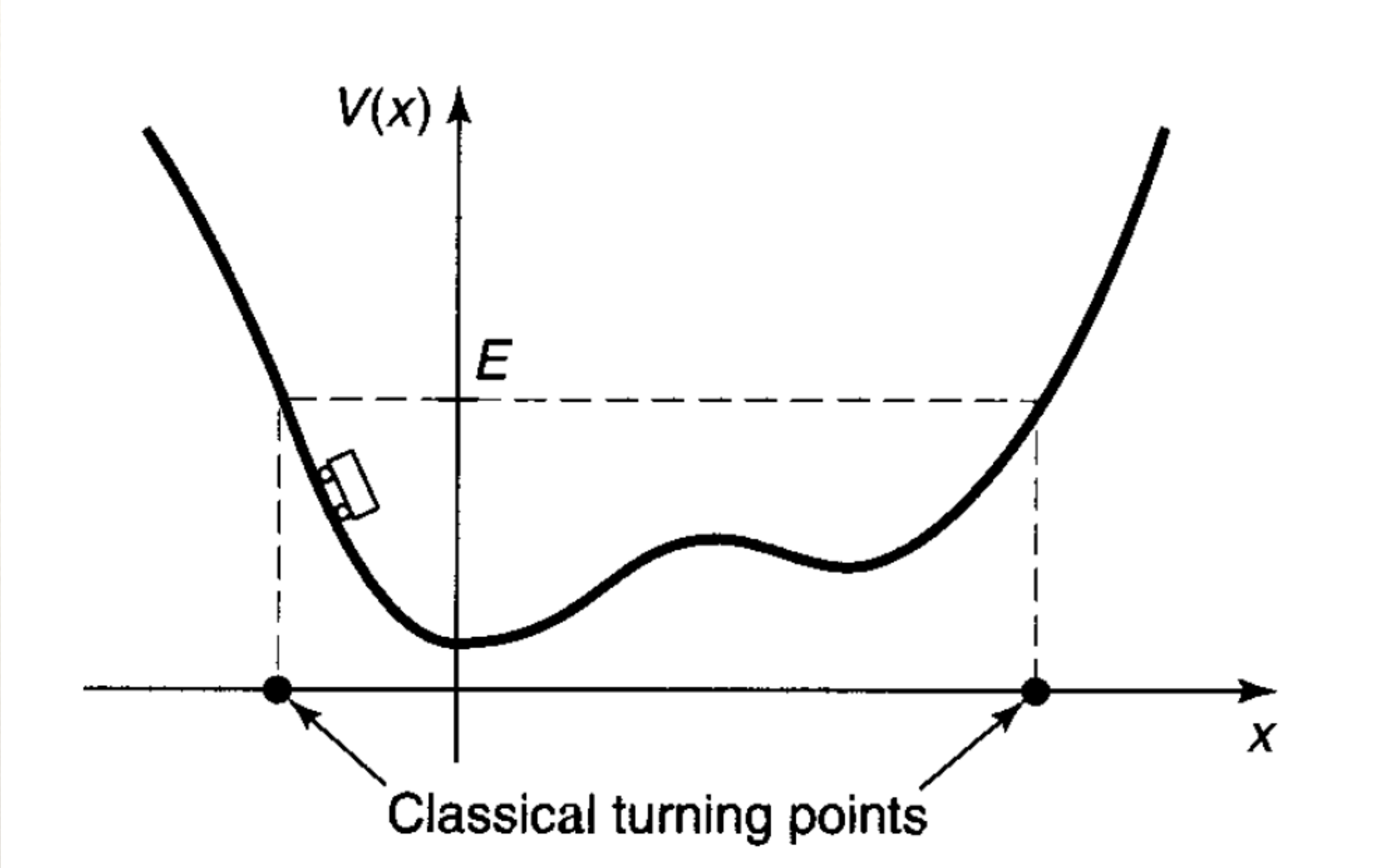
$$\boxed{\psi_n(x) = \sqrt{\frac{2}{a}}\sin\left(\frac{n\pi}{a}x\right).}$$

$$n = 1, 2, 3...$$

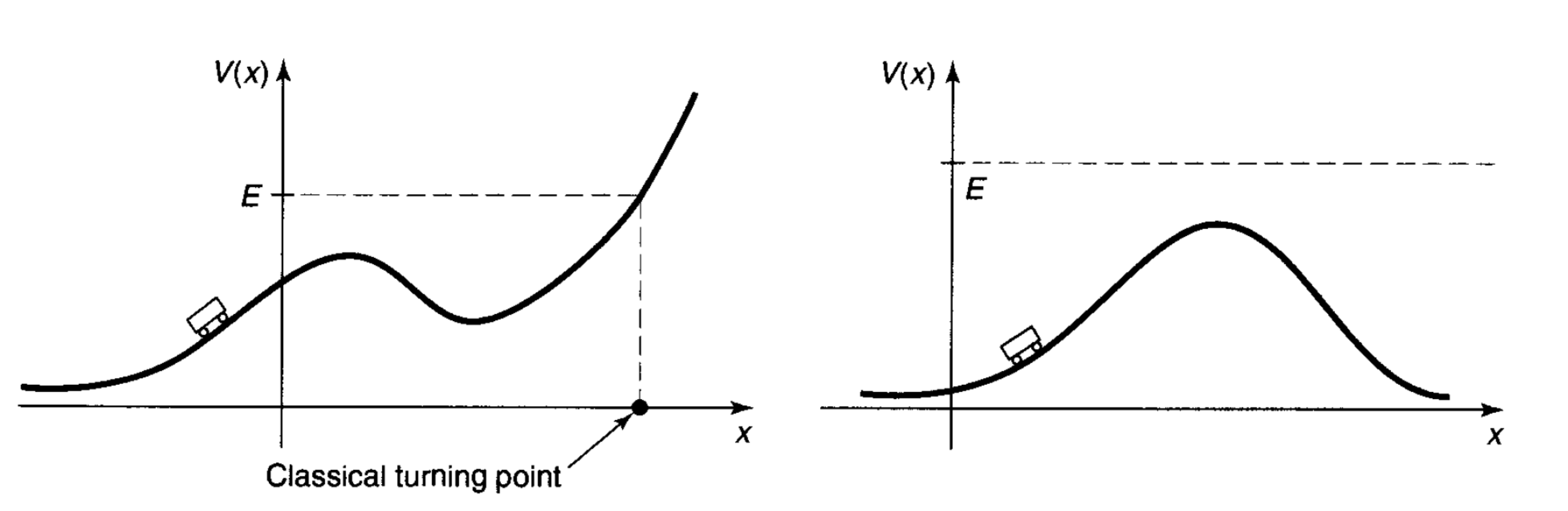
$$\boxed{\Psi(x) = \sum_{n=1}^{\infty} c_n\psi_n(x)}$$

Question: What is the physical significance of this distinction?

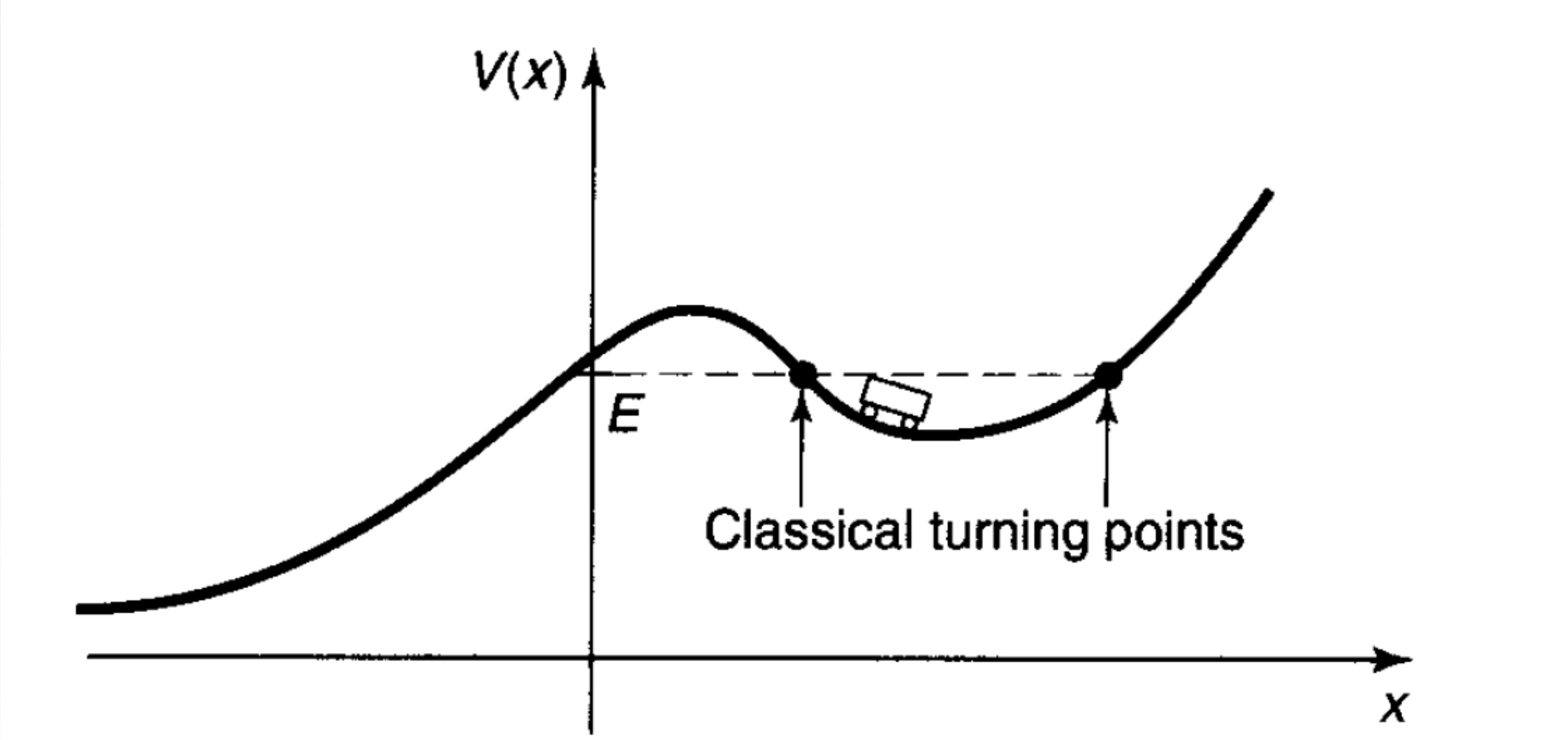
Bound state and Scattering state



Classical bound state



Classical scattering state



Classical bound state or scattering state depending on energy

Quantum bound state/scattering state

Quantum mechanically, a particle can leak through a finite potential barrier: Tunnelling

$$\begin{cases} E < V(-\infty) \text{ and } V(+\infty) \Rightarrow & \text{bound state,} \\ E > V(-\infty) \text{ or } V(+\infty) \Rightarrow & \text{scattering state.} \end{cases}$$

In "real life" most potentials go to zero at infinity.

$$\begin{cases} E < 0 \Rightarrow & \text{bound state,} \\ E > 0 \Rightarrow & \text{scattering state.} \end{cases}$$

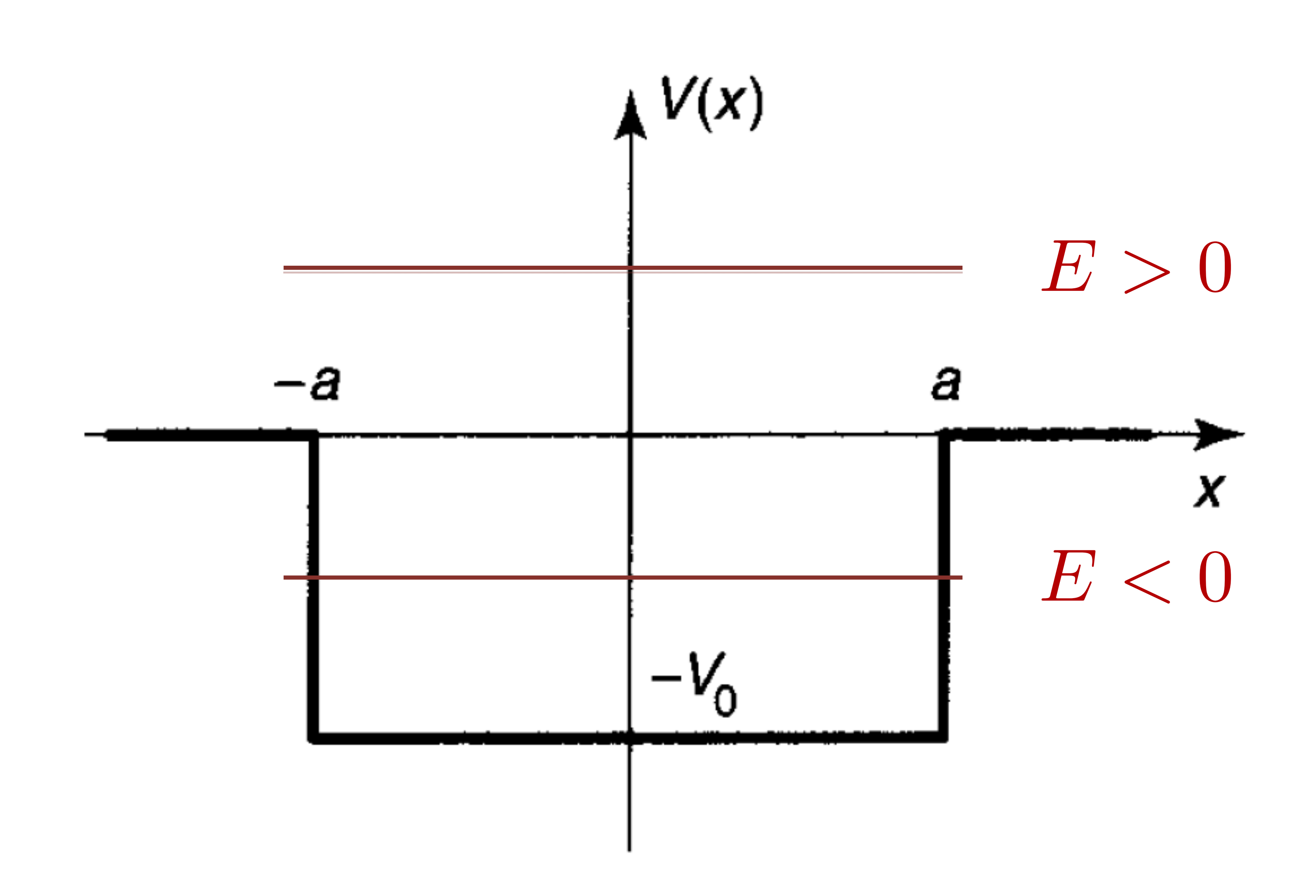
In infinite square well, the potential the potential goes to infinity as x goes to infinity, so they only admit bound state solution.

For free particle, potential is zero everywhere, so they admit scattering state solution only.

Finite Square well potential

$$V(x) = \begin{cases} -V_0, & \text{for } -a < x < a, \\ 0, & \text{for } |x| > a, \end{cases}$$

V_0 is a constant with $V_0 > 0$.



This potential presumably admits both kinds of solutions:

Bound state solution for $E < 0$

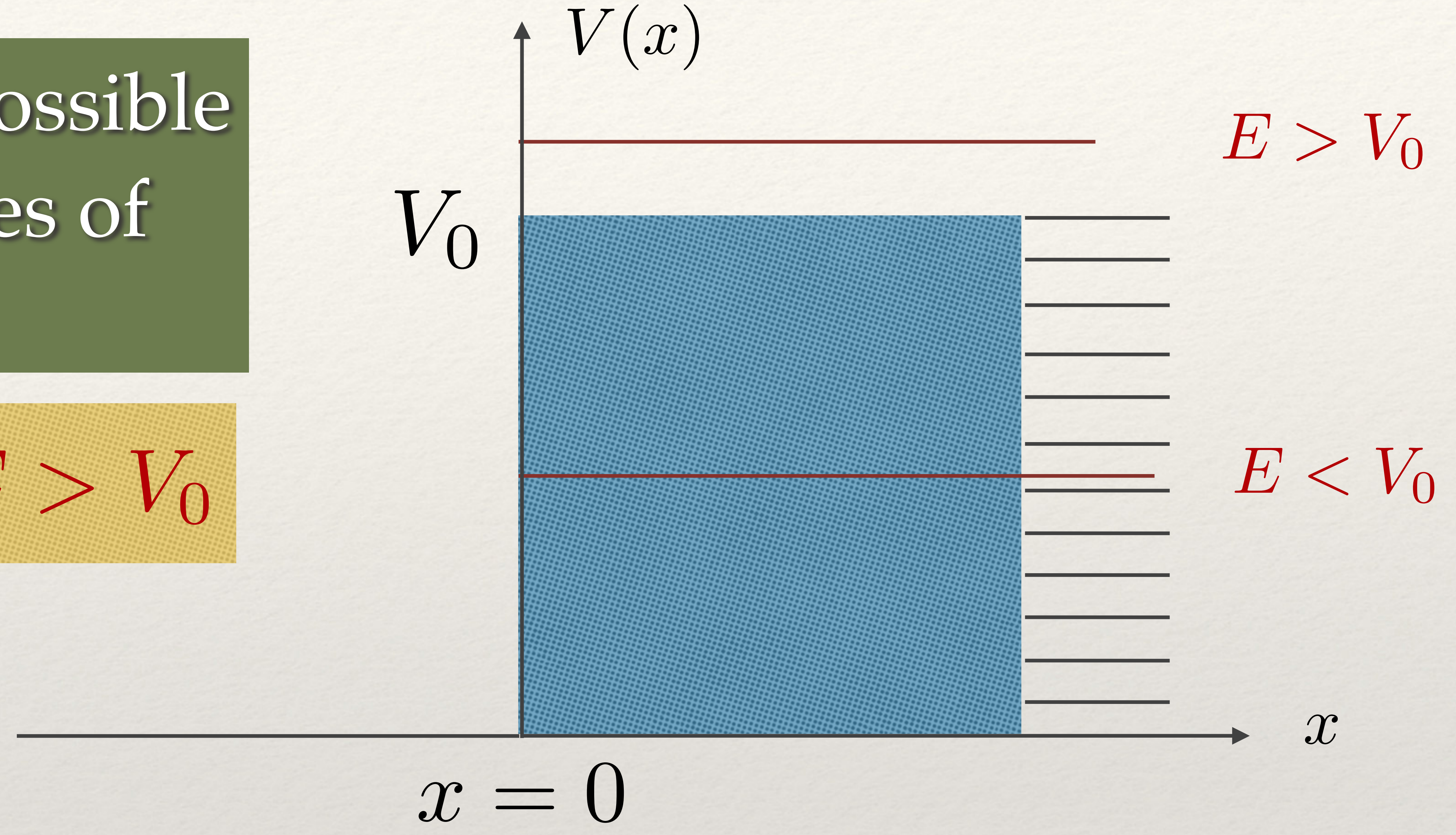
Scattering state solution for $E > 0$

Step Potential

$$V(x) = 0 \qquad x < 0$$
$$= V_0 \qquad x > 0$$

There are two possible energy regimes of interest:

$E < V_0$ $\qquad$ $E > V_0$



We will again use separation of variables to write time independent Schrodinger equation

$E < V_0$

$$x < 0: \quad \frac{d^2\psi}{dx^2} + k^2\psi = 0; \quad k = (\frac{2mE}{\hbar^2})^{1/2}$$

$$x > 0: \quad \frac{d^2\psi}{dx^2} - \kappa^2\psi = 0; \quad \kappa = [\frac{2m}{\hbar^2}(V_0 - E)]^{1/2}$$

Solution:

$$x < 0, \quad \psi(x) = Ae^{ikx} + Be^{-ikx} \quad \psi_I(x)$$
$$x > 0, \quad \psi(x) = Ce^{\kappa x} + De^{-\kappa x} \quad \psi_{II}(x)$$

But, $C = 0$ because $\psi \to 0, x \to \infty \Rightarrow \psi(x)|_{x>0} = De^{-\kappa x}$.

Boundary condition and solution

$$\psi_I(x)|_{x=0} = \psi_{II}(x)|_{x=0} \rightarrow A + B = D$$

$$\frac{\partial \psi_I(x)}{\partial x}|_{x=0} = \frac{\partial \psi_{II}(x)}{\partial x}|_{x=0} \rightarrow ik(A - B) = -\kappa D$$

We obtain scattering state solutions, no discrete energy states, this is in agreement with the classification of potential that we did.

$$A = \frac{1 + i\kappa/k}{2}D, \quad B = \frac{1 - i\kappa/k}{2}D$$

$$\frac{B}{A} = \frac{1 - i\sqrt{(\frac{V_0}{E} - 1)}}{1 + i\sqrt{(\frac{V_0}{E} - 1)}}$$

$$\psi_I(x) = \frac{D}{2}(1 + i\kappa/k)e^{ikx} + \frac{D}{2}(1 - i\kappa/k)e^{-ikx}$$

$$\psi_{II}(x) = De^{-\kappa x}$$

D remains unknown, which determines the amplitude of the eigenfunction. Solutions of any amplitude is allowed!

$$\frac{|B|^2}{|A|^2} = 1$$

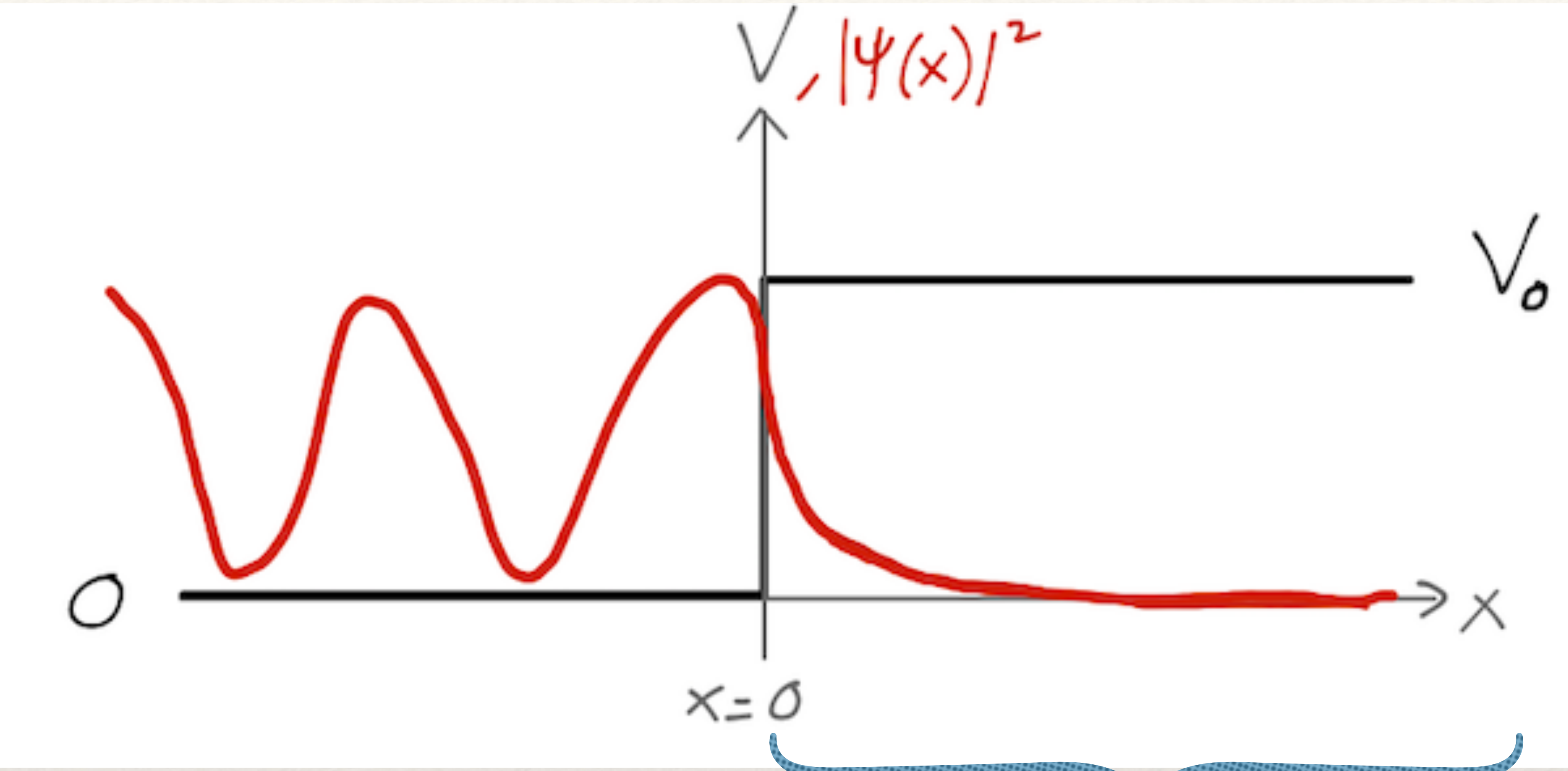
$$R = \frac{|B|^2}{|A|^2} = 1$$

Reflection coefficient

The probability of an incident particle to reflect back with energy less than the barrier is one. That means it is always reflected. This is classically true!

Observations

- For $x < 0 \longrightarrow$ two plane waves moving towards right & left
- $Ae^{ikx} \longrightarrow$ wave of amplitude A incident from left, $Ae^{-ikx} \longrightarrow$ wave of amplitude B reflected from step
- Reflection co-efficient, $R = \frac{v|B|^2}{v|A|^2} = 1 \Rightarrow$ Total reflection



Barrier penetration

◆ For x<0: The probability density of finding the particle:

$$P(x) = \frac{|D|^2}{\cos^2\alpha/2}\cos^2(kx - \alpha/2)$$

Oscillatory behaviour because of the superposition of reflected and transmitted wave

$$\boxed{\alpha = 2\tan^{-1}(-\frac{\kappa}{k})}$$

◆ For x>0: $P(x) = \psi^*_{II}\psi_{II} = |D|^2 e^{-2\kappa x}$

Finite probability in classically forbidden region: Barrier penetration

An appreciable length that it can penetrate: $\Delta x \sim 1/\kappa = \frac{\hbar}{\sqrt{2m(V_0 - E)}}$

$$\Delta p \simeq \frac{\hbar}{\Delta x} \simeq \sqrt{2m(V_0 - E)}$$

$$\Delta E \simeq \frac{(\Delta p)^2}{2m} \simeq V_0 - E$$

Summary

- Separable solutions provide energy eigenstates, However for free particle, there is no such case, they can only be represented by wave packet.

- Solutions are classified as Bound state and/or Scattering state, depending on the kind of potential that particle is subject to.

- Finite potential well admits both bound state and scattering state solutions. Infinite potential well only admit bound state and free particle solutions are only scattering state solutions.

- Step potential provides scattering state solution.

  For step potential we find the reflection coefficient $R = 1$

- The wave function penetrates into the barrier, classically forbidden region even if the energy is smaller than potential- barrier penetration.